Time Emergence: FFGFT and GALG Compared

Decompactification vs. Co-exclusion —
Two Algebraic Paths to Observable Time

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Time Emergence and Gravitational Constant: Correction Proposals for Matzke's GALG Framework

Abstract

This document formulates two correction proposals for Doug Matzke's GALG framework [1, 7], supported by direct quotations and algebraic verification.

Proposal 1 (time): Matzke's co-exclusion primitive generates proto-time through operator change [1] (p. 37), in direct parallel to FFGFT decompactification (unrolling, Doc. 333). Both frameworks share the same three-layer emergence structure. This is not a weakness of GALG but an unspoken parallel that strengthens it.

Proposal 2 (G): The "derivation" of Schwarzschild geometry claimed in [7] "without free parameters" uses $M_{\text{bit}} = m_P \cdot \ln 2/(2\pi)$ as starting point. Since $m_P = \sqrt{\hbar c/G}$ by definition, G is already present in the input. The implicit G agrees with G_{SI} to 0.00% — but as a tautology, not a derivation. The proposal: name this constraint explicitly and leave open the path to a genuine G derivation. In FFGFT, $G = \xi^2 \hbar c/(4m_e^2) \cdot K_{\text{frak}}$ is derived without m_P as input (Doc. 012).

Both proposals are documented through direct quotations from Matzke's publications and verified by Python scripts.

Guide

The basic problem. Observable time is a macroscopic phenomenon. Quantum mechanics and algebraic field theories operate at a level where time is either absent or appears only as a discrete primitive. How does continuous, observable time arise from a timeless algebraic base structure?

What this document shows. Two precise correction proposals for Doug Matzke, supported by direct quotations from his own publications: (1) Time emergence in GALG has the same structure as in FFGFT — this strengthens Doug's framework but has not yet been made explicit. (2) The claimed G derivation in [7] is a tautology — this limits the claim but does not weaken the algebra.

Notation

Symbol	Meaning	Source
Status markers		
[B]	algebraically proved, script available	Doc. 190
[K]	derivable from ξ	Doc. 190
[S]	open bridge, sketch	Doc. 190
FFGFT		
$\tilde{T} = 1/m$	intrinsic time scale of a mode	Doc. 321 [4]
$T^4/\mathbb{Z}_3$	compact 4-torus geometry	Doc. 257 [3]
Decompactification	unrolling of the compact dimension	Doc. 333 [5]
Matzke GALG		
Co-occurrence	addition $a + b$: spatial primitive	[1] p. 37
Co-exclusion	multiplication ab: temporal primitive	[1] p. 37
Proto-time	emergent time order from co-exclusion	[1] p. 14

.1 Time in FFGFT: compactified and emergent

The compactified time scale $\tilde{T} = 1/m$

In FFGFT, each particle of mass m carries an intrinsic time scale (Doc. 321 [4]):

$$\tilde{T} = \frac{1}{m} \qquad \text{(natural units, } \hbar = c = 1\text{)} \tag{1}$$

This time scale lives on the compact geometry $\mathrm{T}^4/\mathbb{Z}_3$. It is mode-specific — each mode has its own time scale, and cross-mode superpositions generate incompatible time scales (Doc. 334 [6]) **[B]**.

Decompactification as the emergence mechanism

The compact dimension is not directly observable. Observable time arises only through unrolling: the transition from compact geometry to macroscopic spacetime (Doc. 333 [5]).

Doc. 333 distinguishes two unrolling types:

- **Type I** (measurement act): lossy, irreversible — wavefunction collapse as decompactification.
- **Type III** (pullback): lossless, reversible — geometric retraction to the compact manifold.

Observable time is a Type-I phenomenon: it arises through the irreversible unrolling act, not from the compact geometry itself.

Core statement FFGFT [B]: Before decompactification, only $\tilde{T} = 1/m$ exists as a mode-bound geometric quantity. Classical, measurable time t is the result of unrolling.

.2 Time in GALG: proto-temporal and emergent

Co-occurrence and co-exclusion

Matzke [1] defines two fundamental primitives for geometric algebra (Ch. 4.2, pp. 36–39):

Co-occurrence (spatial primitive): two states exist simultaneously, expressed as GA addition $a + b$. Direct quote [1], p. 37:

> "Co-occurrence is defined as two or more states 'occurring' simultaneously [...] commutative properties of addition reflects the lack of temporal ordering characteristic of simultaneity."

Co-occurrence is static — no temporal ordering.

Co-exclusion (temporal primitive): an operator changes the state through GA multiplication ab. Direct quote [1], p. 37:

> "Since co-exclusion infers an operator for change and implies temporal sequence, it is considered to be the temporal primitive. Conversely, co-occurrence is labeled the spatial primitive since it is static."

Proto-time emerges from operators

Classical time is not encoded in the GALG framework, but emerges from the interaction of disjoint operators. Direct quote [1], p. 14:

> "Interaction of disjoint finite sets of dimensions (operators and resulting co-exclusions) creates a synchronization-based proto-time from which classical time and energy metrics ultimately emerge."

On the need for a different temporal concept in the quantum domain (p. 12):

> "The quantum domain must have a different kind of primitive temporal framework, if for no other reason than it exists in a completely different spatial environment. Conceptually, this proto-temporal framework is identical to the synchronization tokens and mechanisms used by asynchronous logic."

Core statement GALG: Before emergence, only the algebraic primitive of co-exclusion (multiplication) exists. Classical time and energy arise from the interaction of disjoint operator sets.

.3 The structural parallel

Layer	FFGFT	GALG
Timeless base layer	$\tilde{T} = 1/m$ compactified	co-occurrence (addition)
Temporal primitive	torsion / winding on T^4	co-exclusion (multiplication ab)
Emergence mechanism	decompactification (unrolling)	interaction of disjoint operators
Result	observable time t, mode-bound	classical time + energy

Both frameworks share the same three-layer structure: (1) a timeless algebraic base layer, (2) a temporal primitive that implies sequence without being continuous itself, (3) an emergence mechanism that produces classical, measurable time.

.4 Open question

Is Matzke's co-exclusion the algebraic expression of the same structure that FFGFT describes as decompactification?

Concretely: in FFGFT, unrolling Type I (measurement act, collapse) is an irreversible transition — just as co-exclusion describes a state change through an operator without possibility of reversal (Matzke [1], p. 145: *"The process is irreversible because a multiplicative inverse does not exist for* $(1 \pm a_0)$ *and* $(1 \pm a_1)$*"*).

This parallel — irreversibility as a shared feature — could be the algebraic core of the connection **[S]**.

Verification would require:

1. Explicit construction of the unrolling operator in GALG notation.
2. Proof that co-exclusion + irreducibility reproduces the FFGFT mass formula $\tilde{T} = 1/m$.

.5 Mass and gravitational constant: derived or implicit?

Matzke's Schwarzschild derivation and the implicit G

In his IPI Letters paper 2026 [7], Matzke claims:

> "This paper derives the complete black hole picture — Schwarzschild radius, Bekenstein entropy, Hawking radiation, collapse threshold, and information conservation — from a single algebraic axiom with no general relativity and no free parameters."

The derivation uses $M_{\text{bit}} = m_P \cdot \ln 2/(2\pi)$ as starting point (Landauer at T_{Planck}, then $E = mc^2$). From the collapse condition Compton = Schwarzschild:

$$\frac{\hbar}{M_{\text{coll}}\, c} = \frac{2G\, M_{\text{coll}}}{c^2} \quad \Rightarrow \quad G = \frac{\hbar c}{2\, M_{\text{coll}}^2} \tag{2}$$

with $M_{\text{coll}} = n_{\text{thresh}} \cdot M_{\text{bit}} = m_P/\sqrt{2}$ (numerically verified: 0.00% deviation).

The implicit G from this construction:

$$G_{\text{doug}} = \frac{\hbar c}{2\, M_{\text{coll}}^2} = \frac{\hbar c}{m_P^2} = 6.6743 \times 10^{-11}\ \text{m}^3/(\text{kg} \cdot \text{s}^2) \tag{3}$$

Deviation from experimental value: $0.00\,\%$.

Why this is not a genuine G derivation

The starting point $M_{\text{bit}} = m_P \cdot \ln 2/(2\pi)$ contains m_P, and the Planck mass is defined as $m_P = \sqrt{\hbar c/G}$. Therefore:

$$G = \frac{\hbar c}{m_P^2} \tag{4}$$

is a **tautology**: G enters through m_P, passes via M_{bit} into M_{coll}, and is then extracted again. The result is algebraically consistent **[S]** but not an independent derivation of G.

Matzke himself confirms this implicitly in his IPI mail of 27 Aug 2026:

> "the fact that M_{bit} is directly related to M_{Planck} was a happy coincident based on Planck units consolidation only."

FFGFT: G as genuine derivation

In FFGFT, G is derived from ξ and m_e (Doc. 012 [8]), *without* m_P as input:

$$G = \frac{\xi^2\,\hbar c}{4\,m_e^2} \cdot K_{\text{frak}} \qquad \text{with } \xi = \frac{4}{30\,000},\ K_{\text{frak}} = \frac{74}{75} \tag{5}$$

Here ξ is fixed by the $\mathrm{T}^4/\mathbb{Z}_3$ geometry and m_e by the mass formula, both independent of G. This is a genuine derivation: G is output, not input.

Comparison

Quantity	GALG (Matzke)	FFGFT
Input	m_P (contains G!)	ξ, m_e
Mass	$M_{\text{bit}} = m_P \cdot \ln 2/(2\pi)$	$\tilde{T} = 1/m$, then ξ
Gravitation	$G = \hbar c/m_P^2$ (tautology)	$G = \xi^2 \hbar c/(4m_e^2) \cdot K$
Status	algebraically consistent **[S]**	genuine derivation **[K]**

Both frameworks recover the same numerical value for G, but by fundamentally different routes: GALG through consistency with the Planck definition, FFGFT through independent derivation from geometric parameters.

Note on AI assistance

This document was produced with AI-assisted support (Claude, Anthropic). All direct quotations were taken from Matzke's PhD [1] and verified with page references. The author bears sole responsibility for the content. Methodological principles: A-series, Doc. 190.

References

Bibliography

[1] D. J. Matzke, *Quantum Computation Using Geometric Algebra*, Ph.D. Dissertation, University of Texas at Dallas, 2002. https://www.matzkefamily.net/doug/papers/PHD/phd_matzke_final.pdf
Direct quotes in this doc.: p. 12 (proto-temporal framework), p. 14 (proto-time from co-exclusion), p. 37 (temporal primitive), p. 145 (irreversibility).

[2] D. J. Matzke, *GALG: Geometric Algebra over $\mathbb{Z}_3\mathbb{C}$*, IPI Talk, 26 July 2025. https://informationphysicsinstitute.blogspot.com/2025/07/ipi-talk-doug-matzke-wwwquantumdougcom.html

[3] J. Pascher, *Doc. 257: T$^4/\mathbb{Z}_3$ Geometry and FFGFT*, FFGFT Corpus, ORCID 0009-0000-6518-4064. https://github.com/jpascher/T0-Time-Mass-Duality

[4] J. Pascher, *Doc. 321: SU(3) and $\mathbb{Z}_3$ Emergence; intrinsic time scale $\tilde{T} = 1/m$*, FFGFT Corpus, ORCID 0009-0000-6518-4064.

[5] J. Pascher, *Doc. 333: K_{frak}, Unrolling and the $\tilde{T} \cdot m$ Duality*, FFGFT Corpus, 27 August 2026, ORCID 0009-0000-6518-4064.

[6] J. Pascher, *Doc. 334: Superposition without Time*, FFGFT Corpus, 27 August 2026, ORCID 0009-0000-6518-4064.

[7] D. J. Matzke, *Black Holes from Hyperbits: The Event Horizon as Nilpotent Algebra, Planck Voxels, and High-Dimensional Corner Geometry*, IPI Letters 2026, Vol. 4(3): N1–N22. https://ipipublishing.org/index.php/ipil/article/download/378/201/1385

[8] J. Pascher, *Doc. 012: Derivation of the gravitational constant G from ξ*, FFGFT Corpus, ORCID 0009-0000-6518-4064. https://github.com/jpascher/T0-Time-Mass-Duality